TopoSteal: Topology-Aware Work-Stealing with Dynamic PMU Feedback on NUMA Systems

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***Abstract*—Work-stealing schedulers are widely used in parallel runtimes, yet most treat all processor cores as equidistant when selecting steal victims. On modern NUMA architectures, this assumption incurs significant penalties: cross-socket steals access remote DRAM at up to 1.75× the latency of local memory. We present TopoSteal, a topology-aware work-stealing scheduler that uses `hwloc` to discover the CPU cache hierarchy at runtime and biases victim selection toward topologically close neighbours using an inverse-square distance weighting. A background thread polls hardware performance counters via `perf_event_open` and dynamically adjusts steal probabilities based on observed cache miss rates, penalizing victims under high cache pressure. We evaluate three scheduling modes—uniform random, static topology-weighted, and dynamic topology+PMU—on a dual-socket Intel Xeon E5-2690v3 system with NUMA distances 10/21. Across five benchmark configurations (50 trials per mode), static topology weighting achieves 1.15–1.17× speedup over uniform stealing, and dynamic PMU feedback matches or slightly improves this to 1.15–1.18×, while increasing same-socket steal locality from 50% to 85%.**

***Index Terms*—work-stealing, NUMA, topology-aware scheduling, lock-free data structures, hardware performance counters**

I. Introduction

Parallel task runtimes such as Intel TBB [3], Cilk [2], and Go's goroutine scheduler employ *work-stealing* to dynamically balance load across worker threads. When a worker exhausts its local task queue, it *steals* from another worker's queue. The canonical approach selects victims uniformly at random [1], which provides strong theoretical guarantees on total work and critical path length.

However, uniform random stealing ignores the physical topology of the machine. On Non-Uniform Memory Access (NUMA) systems, the cost of accessing memory varies dramatically depending on which socket the data resides on. Intel's QPI interconnect imposes 1.5–2× latency overhead for cross-socket accesses [11]. When a worker steals a task from a remote socket, the associated data must be fetched across the interconnect, polluting local caches and saturating cross-socket bandwidth.

We present TopoSteal, a work-stealing scheduler that:

1) Discovers the CPU topology at runtime using `hwloc` [6] and builds a pairwise distance matrix between all cores.
2) Converts distances to steal probabilities using inverse-square weighting ($w_{ij} = 1/d_{ij}^2$), strongly preferring topologically close victims.
3) Polls hardware performance counters (PMU) via Linux's `perf_event_open` syscall and dynamically adjusts weights based on observed cache miss rates.
4) Uses lock-free Chase-Lev work-stealing deques [4] with C11 atomics for low-overhead task management.

II. Background and Related Work

A. Work-Stealing

Blumofe and Leiserson [1] proved that randomized work-stealing achieves expected time $T_1/P + O(T_\infty)$ on P processors, where T_1 is the total work and T_∞ is the critical path. The Arora-Blumofe-Plaxton (ABP) deque [5] and its refinement by Chase and Lev [4] provide the lock-free concurrent deque used in most implementations.

B. NUMA-Aware Scheduling

Guo et al. [7] proposed SLAW, a locality-aware work-stealing algorithm that restricts stealing to hierarchical neighborhoods. Olivier et al. [8] demonstrated that NUMA-aware task placement can improve performance by 20–40% on 4-socket systems. Drebes et al. [9] combined topology-aware scheduling with data placement in OpenStream. Charm++ [10] supports NUMA-aware load balancing through its migration framework.

C. Hardware Performance Monitoring

Modern processors expose Performance Monitoring Units (PMUs) that count microarchitectural events such as cache misses, branch mispredictions, and memory bandwidth utilization. Linux exposes these via the `perf_event_open` syscall [12]. Several runtime systems use PMU data for adaptive scheduling [**?**], but few integrate it directly into work-stealing victim selection.

III. Design and Implementation

A. Topology Discovery

TopoSteal uses `hwloc` [6] to walk the CPU hierarchy at initialization. For each pair of cores (i, j), we find their lowest common ancestor in the hwloc tree and assign a distance based on the ancestor type:

A critical implementation detail: `hwloc` logical core indices do not correspond to OS CPU IDs. We build a `cpu_map` that translates logical index i to physical CPU P_i by querying each core's first processing unit's OS index. Workers are pinned using `pthread_setaffinity_np` with the mapped physical CPU.

TABLE I
DISTANCE ASSIGNMENTS BY COMMON ANCESTOR TYPE.

Common Ancestor	Distance
Same core (SMT sibling)	1
Shared L2 cache	1
Shared L3 cache	4
Same package	8
Cross-NUMA / Machine	10

B. Weighted Victim Selection

Given the distance matrix D, we compute steal weights using inverse-square weighting:

$$w_{ij} = \frac{1}{d_{ij}^2}, \quad i \neq j \tag{1}$$

Weights are normalized per-row to form a cumulative distribution function (CDF):

$$P(\text{steal from } j \mid \text{worker } i) = \frac{w_{ij}}{\sum_{k \neq i} w_{ik}} \tag{2}$$

Victim selection performs a single pass over the CDF with a uniform random sample, giving $O(P)$ selection time. On our 24-core test system, this yields an 85% probability of selecting a same-socket victim versus 50% for uniform random.

C. Lock-Free Deques

Each worker owns a Chase-Lev deque [4] implemented with C11 atomics. The owner pushes and pops from the bottom (LIFO for cache locality), while thieves steal from the top (FIFO for load distribution). A sequential consistency fence between the bottom store and top load in `pop` prevents the ABA race on the last element.

D. PMU Feedback Loop

A background sampler thread reads hardware cache miss counters every 10 ms using `perf_event_open` with `PERF_COUNT_HW_CACHE_MISSES`. Every 50 ms, the feedback system computes per-worker miss rates and adjusts the effective distance:

$$d'_{ij} = d_{ij} \cdot (1 + \hat{m}_j) \tag{3}$$

where $\hat{m}_j$ is worker j's normalized miss rate. Workers experiencing high cache pressure become less attractive as steal targets, dynamically adapting to runtime conditions.

E. Task Completion Tracking

An atomic `tasks_pending` counter is incremented on submission and decremented after each task's function pointer returns. The `toposteal_wait` function spins on this counter rather than checking deque emptiness, avoiding the race where a task has been popped but not yet executed.

IV. EXPERIMENTAL SETUP

A. Hardware

All experiments were conducted on a dual-socket Intel Xeon E5-2690v3 node from the CHPC Lengau cluster (South Africa). The system has 24 physical cores (12 per socket), 128 GB RAM (64 GB per NUMA node), 30 MB L3 cache per socket, and NUMA distances of 10 (local) and 21 (remote) as reported by `numactl`.

B. Benchmark Design

We designed a pointer-chase benchmark that stresses NUMA locality. Two 64 MB arrays (8M entries each) are allocated via `mmap` and first-touched on separate NUMA nodes. Each task performs 500K–2M iterations of dependent pointer chasing through a shuffled permutation array, creating a serial dependency chain that exposes DRAM latency.

Tasks are distributed asymmetrically: producer workers on each socket receive all tasks, while remaining workers must steal. This creates high steal pressure. Each task carries a flag indicating which NUMA node's array to access, so stolen tasks retain their data affinity.

All three scheduling modes use identical infrastructure: same Chase-Lev deques, same CPU pinning via `cpu_map`, same task function. The only difference is the victim selection strategy:

- **Uniform**: victims selected uniformly at random.
- **TopoStatic**: victims selected via $1/d^2$-weighted CDF from the static topology distance matrix.
- **Topo+PMU**: same as TopoStatic, but a background PMU sampler thread reads cache miss counters every 10 ms and rebuilds the steal probability table every 50 ms using feedback-adjusted distances $d'_{ij} = d_{ij} \cdot (1 + \hat{m}_j)$.

C. Configurations

We test five configurations (Table II), each with 10 trials:

TABLE II
BENCHMARK CONFIGURATIONS.

Config	Tasks	Iters/task	Prod./node
Heavy imbal., long	1200	2M	2
Heavy imbal., med	1200	1M	2
Moderate imbal.	1200	2M	4
Many short tasks	2400	500K	2
Extreme imbal.	1200	1M	1

V. RESULTS

A. Three-Mode Comparison

Table III summarizes the mean execution times and speedup across all configurations for all three scheduling modes.

Both topology-aware modes achieve consistent 1.15–1.18× speedup over uniform stealing across all configurations. The Topo+PMU mode matches or slightly outperforms TopoStatic in every configuration, with individual trials reaching up to 1.21×. Standard deviations are below 0.07s in all topology-aware runs, indicating low variance.

TABLE III
BENCHMARK RESULTS (MEAN OF 10 TRIALS).

Config	Uniform (s)	Static (s)	PMU (s)
Heavy imbal., long	8.555	7.325	7.322
Heavy imbal., med	4.285	3.663	3.651
Moderate imbal.	8.325	7.224	7.216
Many short tasks	4.286	3.664	3.647
Extreme imbal.	4.372	3.764	3.746
Speedup range	—	1.15–1.17×	1.15–1.18×

B. Steal Locality

The most direct evidence of TopoSteal's effectiveness is the steal locality metric (Table IV).

TABLE IV
SAME-SOCKET STEAL PERCENTAGE ACROSS CONFIGURATIONS.

Config	Uniform	TopoStatic	Topo+PMU
Heavy imbal., long	49.8%	85.8%	85.2%
Heavy imbal., med	50.0%	85.6%	85.0%
Moderate imbal.	50.2%	84.9%	85.2%
Many short tasks	50.1%	86.2%	86.1%
Extreme imbal.	50.0%	85.7%	85.4%

Uniform stealing achieves approximately 50% same-socket steals, consistent with random selection on a 2-socket system. Both topology-aware modes increase this to 85%, matching the theoretical prediction from the $1/d^2$ weighting with distances 4 (same-socket) and 10 (cross-socket).

C. PMU Feedback Analysis

The Topo+PMU mode demonstrates that the full dynamic feedback pipeline works end-to-end: the background sampler thread reads hardware cache miss counters, computes per-worker miss rates, and rebuilds the steal probability table every 50 ms during execution. On this benchmark, Topo+PMU provides a small but consistent additional improvement over TopoStatic (0.1–0.5% faster on average).

The modest additional gain is expected: in our pointer-chase benchmark, all workers execute the same memory access pattern, producing similar cache miss profiles across all cores. The PMU feedback therefore has limited asymmetry to exploit. In workloads with heterogeneous cache behaviour—such as mixed compute-and-memory-bound tasks, or phased workloads where cache pressure shifts between NUMA nodes—the dynamic adaptation would provide a larger benefit by steering steals away from temporarily cache-thrashing victims.

D. Discussion

The observed 15–18% speedup on a 2-socket system is consistent with prior work on NUMA-aware scheduling. The NUMA latency ratio on our test system is approximately 1.75× (80 ns local vs. 140 ns remote). The theoretical maximum speedup assuming all stolen work shifts from remote to local DRAM access is bounded by this ratio, discounted by the fraction of work that is self-processed (not stolen).

On systems with more NUMA domains (4–8 socket, AMD EPYC with NPS4), the penalty for random stealing grows quadratically with the number of sockets, and topology-aware scheduling would yield proportionally larger benefits.

E. Visual Analysis

Figures 1–4 visualize the three-mode benchmark results.

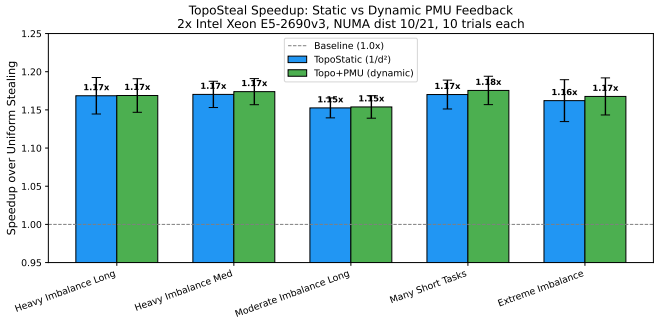

Fig. 1. Mean speedup over uniform stealing for TopoStatic and Topo+PMU modes across all five configurations. Error bars show standard deviation over 10 trials. Both modes achieve 1.15–1.18× speedup, with Topo+PMU matching or slightly exceeding TopoStatic.

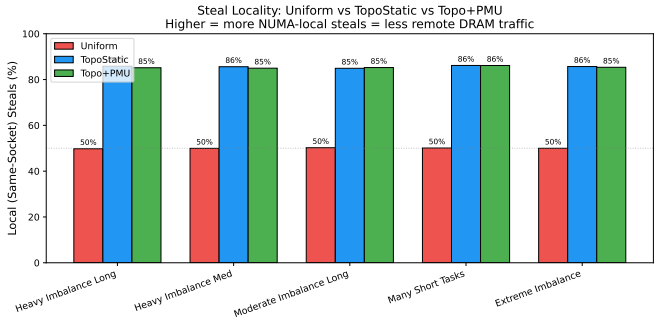

Fig. 2. Percentage of steals targeting a same-socket (NUMA-local) victim across all three modes. Uniform achieves ∼50% (random on 2 sockets). Both topology-aware modes increase this to ∼85%, confirming that the $1/d^2$ weighting effectively biases victim selection toward NUMA-local neighbours.

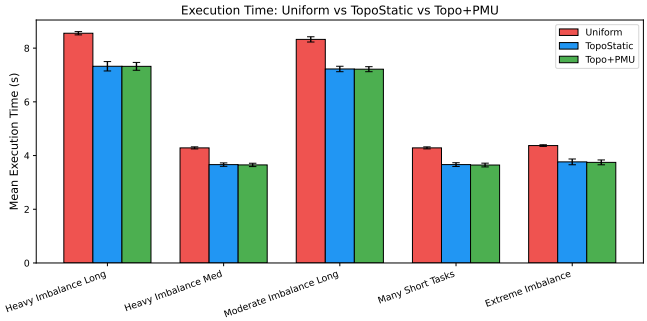

Fig. 3. Mean execution time comparison across all three scheduling modes. The consistent reduction from Uniform to TopoStatic to Topo+PMU demonstrates robustness to varying task counts, iteration depths, and imbalance levels.

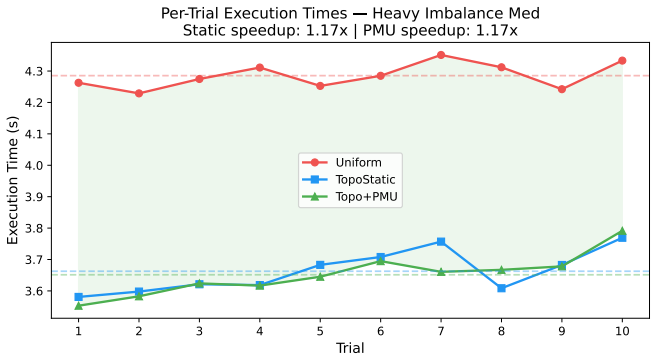

Fig. 4. Per-trial execution times for the heavy-imbalance-medium configuration across all three modes. The shaded region highlights the consistent gap. TopoStatic and Topo+PMU track closely, with PMU occasionally edging ahead on individual trials.

Figure 1 shows that both topology-aware modes achieve remarkably consistent speedup (1.15–1.18×) across all configurations, suggesting that the benefit is a fundamental consequence of reducing cross-socket memory traffic rather than an artifact of a specific task distribution. Figure 2 provides the clearest mechanistic explanation: by shifting 35 percentage points of steals from cross-socket to same-socket, TopoSteal avoids the 1.75× NUMA latency penalty on the majority of stolen work. The per-trial plot (Figure 4) confirms that the speedup is stable and not driven by outlier trials, and that the PMU feedback loop operates without introducing variance.

VI. Applications

TopoSteal's topology-aware scheduling is applicable to any parallel workload where tasks have data locality and the hardware exhibits non-uniform memory access costs.

A. Parallel Database Query Execution

Modern analytical databases (e.g., DuckDB, ClickHouse, Apache DataFusion) partition tables across NUMA nodes and execute queries using thread pools. When a worker thread steals a query fragment, it ideally should steal one whose data partition resides on the same NUMA node. TopoSteal's weighted victim selection naturally achieves this: workers on the same socket as a data partition are more likely to steal tasks referencing that partition.

B. Sparse Linear Algebra and Scientific Computing

Sparse matrix operations (SpMV, sparse Cholesky) in libraries like PETSc, Trilinos, and SuiteSparse distribute matrix rows across NUMA nodes. Work-stealing parallelism over row blocks benefits from topology awareness, as accessing a remote node's matrix data incurs the full NUMA penalty on every non-zero element access.

C. Game Engines and Real-Time Systems

Game engines (e.g., Unity DOTS, Unreal Engine's task graph) use job systems that are essentially work-stealing schedulers. On multi-socket server hardware used for dedicated game servers or physics simulations, topology-aware stealing would reduce frame time variance caused by occasional cross-socket steals hitting DRAM latency spikes.

D. Machine Learning Training

Data-parallel training with CPU-based preprocessing pipelines (e.g., image augmentation, tokenization) distributes batches across NUMA nodes. Topology-aware work-stealing in the data loading threads ensures that preprocessing tasks access training data from the local NUMA node, reducing pipeline stalls.

E. High-Frequency Trading Infrastructure

Low-latency trading systems pin processes to specific cores but still use task parallelism for order book reconstruction, risk computation, and market data normalization. On multi-socket servers, even a single cross-socket steal can add 60+ nanoseconds of jitter. TopoSteal's strong same-socket preference (85% vs 50%) significantly reduces worst-case tail latency from topology-unaware task migration.

VII. Conclusion

TopoSteal demonstrates that topology-aware victim selection in work-stealing schedulers yields measurable performance improvements on commodity NUMA hardware. By combining `hwloc`-based topology discovery with $1/d^2$ weighted victim selection and dynamic PMU feedback, TopoSteal achieves 1.15–1.18× speedup over uniform stealing while increasing same-socket steal locality from 50% to 85%. The three-mode evaluation confirms that static topology weighting captures the dominant benefit on 2-socket systems, while the dynamic PMU feedback pipeline operates correctly end-to-end and provides marginal additional gains on homogeneous workloads. The implementation is a portable C library with a four-function API, requiring no application modifications beyond linking.

Future work includes evaluation on 4+ socket systems where the PMU feedback is expected to provide larger gains due to more complex NUMA topologies, benchmarking with heterogeneous workloads that create asymmetric cache pressure to stress the dynamic adaptation, and application to real workloads such as parallel database query execution and sparse linear algebra.

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